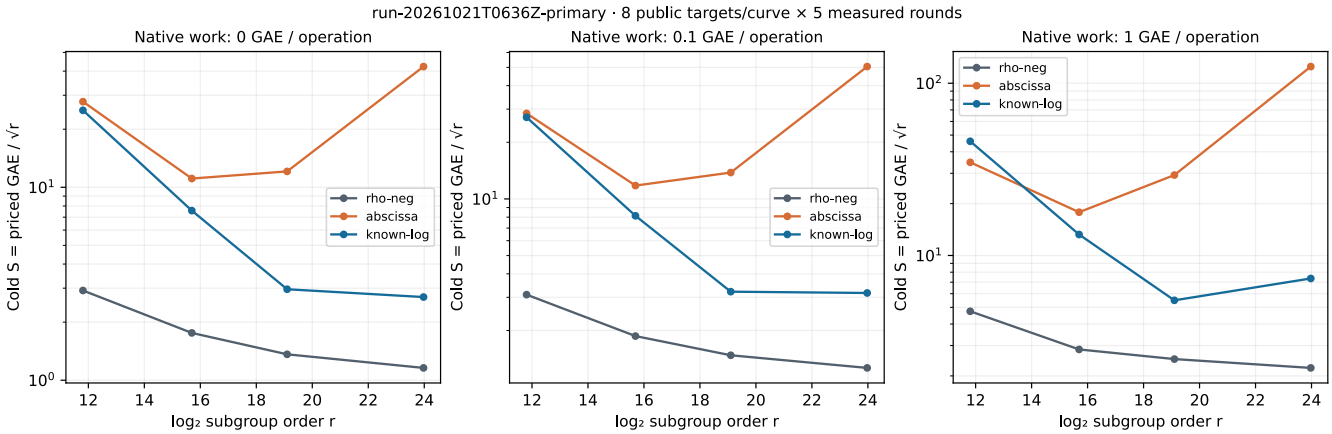
Arbitrary-target known-log recovery

2026-10-21 · Frozen synthetic prime-field study · Same-host validation

The target-blind known-log two-sum method recovered public target logs without receiving their values. Its cold counted cost is lower than the specified abscissa IC pipeline at the three larger sizes in both primary and holdout panels. **Negation rho remains cheaper at every size.** The smallest comparison changes ordering across the native-cost band. Independent validation and complete native-cost calibration remain open.

Cold measured costs

Eight public targets per curve, five measured rounds and one warmup per arm; 32 unsigned base pairs (64 signed points). Setup is rebuilt and charged for each target and round. S = GAE / √r, with GAE counting group additions and doublings. The figure displays native per-curve means; it fits no exponent.



log₂ r	Abscissa S(0)	Known-log S(0)	Rho S(0)	Abscissa / known-log at prices 0, 0.1, 1
11.807	27.7819	25.0571	2.9177	1.1087, 1.0488, 0.7550
15.697	11.1004	7.5688	1.7587	1.4666, 1.4472, 1.3476
19.101	12.0860	2.9616	1.3624	4.0810, 4.2953, 5.3357
23.974	42.2267	2.6984	1.1584	15.6487, 15.9587, 16.9821

At 24 bits the holdout abscissa / known-log ratios are 16.6589, 17.0190 and 18.2168. The primary price-zero candidate / baseline ratio is 0.063903, with 95% target-cluster bootstrap interval [0.044228, 0.087908]. This interval is from ecbench; it is distinct from the pricing-sensitivity band. The matched rho / candidate S ratios are 0.4293, 0.3998 and 0.3025, so rho remains cheaper.

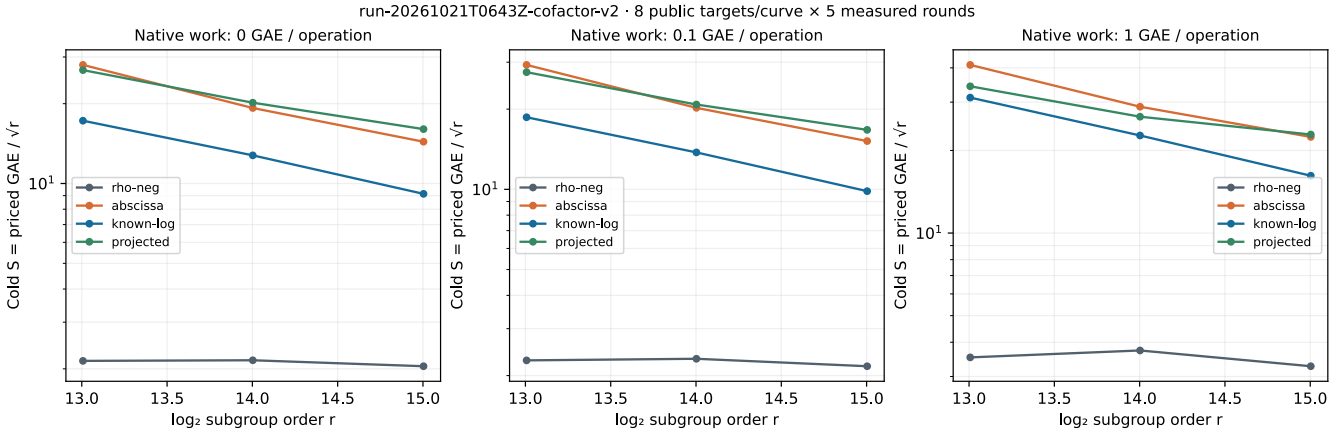
The comparator is the unchanged prime-abscissa B32 two-sum MITM pipeline with incremental Gaussian elimination. It is the specified baseline, not an established globally best IC configuration. The candidate is generic table search with known base logs, not evidence of a new algebraic trapdoor.

Source: study README, protocol, sealed sessions, native summaries and paired comparisons. The README lists exact registered curve identities, costs and uncertainty.

Controls, collisions and validation

Preparation accepts no target. A checked full-point decomposition gives one useful equation in the unknown target log; unknown base logs and matrix work are zero. The runner independently checks the answer. Baseline final matrix rank, dependent rows and elimination work are retained; a rank-growth time series is not implemented.

Generator panels use G, [2]G and [7]G. Transporting the base by the inverse multiplier preserves its point-set hash exactly. A fixed-query-seed exhaustive test checks scalar conversion. Cross-generator harness seeds differ, so those timings establish no generator-dependent hardness change.



The corrected cofactor panel fixes p = 65519. With 1,024 nonidentity pair attempts, h = 2, 4 and 8 yield 993, 957 and 911 distinct signed sum classes, and 31, 67 and 113 colliding attempts. The respective subgroup orders are 32,843, 16,411 and 8,231. Changing both the subgroup and curve model prevents a causal cofactor conclusion. Negation is folded; other symmetry contributions are not isolated. These are not NIST P-192/P-224 recovery results.

Across eight sessions, 4,484 executions comprise 4,340 verified recoveries and 144 retained projected-control setup errors. All 3,610 verified measured replays are identical on the measuring host. Totals include warmups, A/A and the corrected rerun; they are not counts of distinct targets. The original cofactor control rejected scan_cap = 65536 at p = 65519. The separately frozen version-two control uses 65518; no failed record was replaced.

Counted A/A ratios equal 1 with interval [1, 1]. Wall comparisons admit no L2 pairs. The host is virtual Linux x86_64 on AMD EPYC 9V74, five visible logical CPUs; one thread, shared CPU, 120-second timeout and 4 GiB address-space limit. Raw RSS and wall values are descriptive. Pricing all emitted native operations at 0, 0.1 and 1 GAE is a sensitivity model, not complete machine calibration.

Library validation: 4,478 tests passed, 119 ignored; 17 ecbench integration tests and three example tests passed. CI-toolchain Rust 1.98 clippy passed. Independent host replay, independent headline recomputation, strongest external comparison and actual multi-target amortization are still missing. No validated leaderboard or benchmark-index row is added, and no asymptotic claim is made.

Next: independently replay and validate these frozen records, calibrate native work, then compare a predeclared B sweep with optimized negation BSGS and the strongest rho on fresh holdouts. See REQUIREMENTS.md, verification/study-audits.json, and problems/known-log-recovery-cost-and-validation.md for the exact open gates.